Changelog

Feature notes in the order they were added (oldest first). The README describes the app as it is today.

- Tree and building heights now update on their own as soon as a measurement has a result (no extra button). Every My Garden card (Time of day, Light probe and more) can be hidden or shown, and map tool rows like "Front door wrong?" fold up. Found objects live in their own movable, folded section, and an "Identified objects" item under At a glance opens it. Scans of your top photo run quietly instead of popping up at sign-in.
- New Normal view with no sun, water or soil layers. Each view shows its own details right under the map: sun and shade for the sun views, water flow for Water, and a button for every bed and growing area in Soil zones. Any section, including At a glance, can be dragged (or moved with the arrow keys) under the map. The Top photo box sits right under the map and folds up.
- Setup: Live Compass is at the top of the line-up screen, and you can turn the house by dragging a finger around the dial's edge.
- Balcony is a new growing place. Balcony and window growers answer which floor they're on, which way it faces (or use the live compass), what's above it and whether walls or a solid railing block sun. The app never asks for an apartment number.
- Apartments: balcony and window growers enter their apartment's size (width and depth, or just the total square feet) instead of a lot. The apartment is drawn at that size, its height comes from the floor it's on, and the balcony or window sits on the side it faces.
- 3D garden (My Garden → 3D garden): the real yard in 3D, including the house, fences, trees, beds, pots and apartment balconies at their floor height. Drag a date slider or press "Watch it grow" to see plants grow, bloom, fruit and die back with the frost, with sun and shadows for the time of day. Side lists show what's in bloom, what's ready to pick and what to do this month. Pick a plant and tap a bed, pot, balcony or the ground to plant it, or tap a plant to move it, change its planting date or remove it. "Save garden file" exports the layout and plants (no photos, notes or address) to send to a client, who opens it with "Open a garden file". Built with plain WebGL, so nothing loads from other sites. The plant library has "Build my garden in 3D" and "Place in 3D garden" buttons.
- Apartment lookup: when an apartment grower types their address, their building's outline appears from OpenStreetMap. They tap the side their home is on (no unit number needed). Nearby buildings come in with heights from the map, or guessed, and can be unticked or corrected. In 3D, the balcony sits on the right wall of the real building.

- Floor plan: upload a floor plan, set the scale from total square feet or one known wall, trace the home, then add each balcony or patio by tapping its corners. Sizes can be adjusted. The app works out which way each one faces, and each becomes its own growing space. The plan image is never saved.
- Look from my balcony: tilt-and-compass readings (or typed angles) of what blocks the sky in each direction. Measured directions replace the map's guesses in the balcony's sun hours.
- My Garden page reorganized: Views sit at the top, with the map under them and the Time of day slider to its right. At a glance lists Overview, Outlines, Where you're growing, Weather, Sun & shade, Soil health, Water flow, Plants in your garden, House photos and Garden notes. The add buttons moved out of the top toolbar into those sections: + Bed, Container and Greenhouse under Where you're growing; + Tree under Plants; + Fence, Neighbor's house and Shed, garage or structure (with height boxes) under Outlines; water tools under Water flow; + Light sensor under Sun & shade. The Normal view shows an overview of everything under the map. Sun & shade now shows which beds are in sun at the chosen time, and Sun hours shows the all-day numbers.
- At a glance tidy-up: House orientation (dial and turn buttons) sits under Overview. Outlines lists the lot and every growing place (front yard, back yard, side yards and more) with Select and width/depth boxes, plus Front door wrong?, the growing-area outline switches, and fences, neighbors' houses and structures. Look ahead moved under Plants in your garden. The Identified objects box shows only when opened from At a glance and has a Close button. Zoom and Undo/Redo moved into the Top photo box under the map.
- Trees: each tree (added, or confirmed from the photo scan) has "How full is it?" with Bushy, Medium, Airy and Bald options, and "Leaves in winter" with drops leaves or evergreen. These are also quick picks under Plants in your garden. Leaf-dropping trees count as bare branches (about a quarter of the shade) outside the estimated leaf season, worked out from your frost dates, in sun hours, shade maps, balcony sun and the 3D view. Balcony sky readings have a Leafy tree tick for each direction, so a tree blocks sun only while it's in leaf.
- Measure objects (At a glance) lists every tree, bush, building and fence with its height. Each has a  Measure button that opens the height tool, and items needing a height are outlined in red on the map. Sun views list Trees & bushes under the map with Type (tree or bush), How full is it?, leaves in winter, a Leaf density slider and Measure buttons. Identified objects: the selected one stays bright while the others fade, and its circle can be dragged to move or resized by its edge handles (buildings by their corners). Opening Plants in your garden highlights beds and growing areas on the map with their plants and lists them under the map, along with measured trees and bushes. Map: two fingers pinch to zoom and drag to move. Setup dial: two fingers twist to turn the photo. Units: lengths in inches, feet, yards or meters, and areas in sq ft, sq yd, sq m or acres

(Overview and Outlines).

- Photo lock: "Done, keep the photo here" on the setup dial and under the map, so the photo stays put once it lines up. Tree height tilt has an Automatic mode that works out how the phone is held (screen up or pointing the back), which fixes near-zero readings on an upright iPad, and it warns about very low angles. Identified objects opens right under the map and can be dragged anywhere. "Missed one?" buttons (+ Tree, Bush or Building) add an object by tapping the map, then you drag and size it. A selected tree's canopy can be resized on the map by dragging its white dots.
- My Garden map has a 2D map / 3D view switch: plot beds in 2D and check them in 3D, with Trees / See-through / No trees. The 3D garden lists every plant with Remove one and Remove all, and you can add a plant bed, rain garden, rock, pebbles or rain barrel by tapping the ground. 3D now shows water marks, pooling, downspouts, the rain gauge, rain barrels, rain gardens, rocks and pebbles. Rain gardens, rocks and pebbles are also new objects on the 2D map (under Where you're growing). The plant library adds Trees, fruit & shrubs (maples, white pine, redbud, serviceberry, apple, peach, fig, hardy banana, blueberry, raspberry, blackberry, elderberry and more), more edibles and herbs (bell and hot peppers, zucchini, cucumber, peas, fennel, rosemary, sage, oregano, dill, cilantro, chives, lavender) and Groundcovers & helpers (clovers, creeping thyme and phlox, wild strawberry, switchgrass, little bluestem, goldenrod, bee balm, anise hyssop, mountain mint, zinnia, comfrey). Benefit filters cover pollinators & butterflies, stopping erosion, groundcover, building soil, rain gardens and birds.
- Map types above the map: 2D map, Lot map, Photos and 3D view. Lot map puts your setup map screenshot under the yard, lined up automatically, or uses one you add in the new Lot map photo slot. Its photo controls and hand mode adjust the lot map. 🖐 Move the map lets you drag around without moving any objects. Plants from the 3D garden show as plant icons on the 2D maps (Show on map → Plants). Photos lets you pick a front, back, side or extra photo and place drawn plants on it: tap to add, drag to move, size with the top handle or slider, and remove. Plants show their season for the chosen date, or full grown. Placements are saved with the garden.
- Setup line-up fixes: the street side of the lot is now worked out from the way the front door faces, so tapping the lot corners in any order gives the right width and depth. While lining up the top photo, the two house corners to tap are marked 1 and 2 on the lot map. Under the photo, a "Not lining up?" row moves, turns (2° or 90°), resizes or flips the outlines, and two fingers on the photo drag, pinch and twist them.
- Compass accuracy: "Check it with the sun". Point the phone's top edge at the sun or along your shadow, and the app works out the compass error from the sun's exact position, then corrects every reading on that device. It warns when the phone is held upright and when the compass disagrees with your address map by more than 25°, and gives tips (hold it flat, step away from metal and cars, wave it in a figure 8).

- Balconies: the side-wall setting now asks which side the wall is on (left or right as you look out), so morning or afternoon sun is counted correctly for corner balconies that are open on one side.
- Townhouses: pick "Townhouse" (end unit with a neighbor on the left or right, or middle unit) in setup or under Overview, and the attached homes, plus optional privacy fences between back patios, are added to the map so they shade your side yards correctly. Choosing "Row house yard" starts as a middle unit.
- Identified objects: "Review identified objects" responds right away (it shows "Opening…" and jumps straight to the list without being hidden under the tabs), and the list is in two columns on wider screens.
- Sun check: tap any spot in Sun & shade or Sun hours to see where the sun is right now, which way shadows point, and exactly what is shading that spot (house, a tree, a neighbor, or the tree-line setting). If it doesn't match your yard, "Fix it with my shadow" turns your map to match: stand in the sun, tap where you stand and where your shadow ends.
- One garden everywhere: plants added to a bed or container (from Beds & plants or the Plant library) appear on the 2D map, in the 3D garden and on matched photos automatically; removing them anywhere updates the bed list.
- Match a photo to your map: in Photos, tap where you stood and what you faced, pick the lens (0.5×, 1×, 2×, 3×) and set the horizon line. Guide lines show your house, beds and trees on the photo so you can line them up. Once matched, your real plants show on the photo and plants you add there go into your 2D map and 3D garden.
- New plants: pear, pawpaw, Raspberry Blast, jalapeño, French marigold and African marigold (blueberry, peach, tomato, parsley, fennel, creeping thyme and strawberries were already in the library).
- Direction lock: a 🔒 button by the direction dial (setup and Overview). When locked, dragging the dial, the turn buttons and the compass can't change which way your house faces. It locks itself after you use the compass or the shadow fix.
- Street side: the lot map marks the street edge with a coral "Street" line. If your lot comes out sideways (width and depth swapped), press "Street is on another side" on the lot map or in the "Not lining up?" box; the lot re-measures and the front direction follows.
- Type of home now includes Apartment or condo. Under Overview (and in setup) you can tick "A deck", "A balcony" and "A patio"; each gets its own outline to drag and measure like a greenhouse. Decks have a height off the ground, balconies a floor.
- Stories: set how many floors your house, townhouse or apartment building has (and which floor you live on). The 3D view draws the building at that height with a line at each floor, and balconies sit on the right floor.

- Compass: readings only count when the phone is flat, steady and the compass reports good accuracy; "Set my front" waits until the reading stops wobbling.
- Compass: fixed iPad/iPhone readings that could come out turned around in landscape, cleared old compass corrections, and added "Type the number from your phone's Compass app" in setup and Overview (sets and locks your front).
- Apartments and townhomes: setup now needs a map screenshot of the lot. Tap Building to trace the whole building, Lot / grounds for the property, then "Your apartment" or "Your townhome" to outline your own home on the building. Your home's size comes from that outline; the rest of the building is placed around it (attached townhomes on each side, or your apartment building).
- Mark what's already there: on a photo matched to your map, switch to "Mark what's already there", pick a tree, bush or plant and tap where it meets the ground. Drag its yellow dot to its top for the height and set the width. It appears on the 2D map, in 3D and in the sun and shade. Tick "Plan to remove" and turn on "Show my yard after removals" to preview the yard, sun and shade without it.
- Plants go straight onto the map: "+ Plant on the map" (Plants in your garden) and "Place on my map" (Plant library) take a plant and a quantity, then a tap on the 2D map. Plants appear in 3D in the same spot; drag them on the 2D map to move them. Adding to a bed from the library also takes a quantity.
- Duplicate beds (and containers, rain gardens, pebbles, sand, mulch, rocks, barrels and more) from their selection card.
- Lot size can be changed in Outlines even after it was plotted; an irregular lot outline stretches to the new size.
- New: In-ground garden as a place you grow; sand; mulch (wood, straw or shredded leaves) as patches on the ground, on top of beds, or as a ring around trees.
- Needed improvements (At a glance): beds and trees that could use mulch, puddles that could become rain gardens, downspouts without barrels and planned removals, with a rough estimate of water mulch keeps in the soil.
- At a glance sections can be moved up and down with the ▲ ▼ buttons; the order is remembered.
- Fixed: plants added to beds sometimes landed in the wrong spot in 3D.
- Site report (At a glance): fill in Prepared for, Prepared by, an optional site address (never saved) and notes, then "Make the report" for a client-ready site assessment: summary, site facts, map, sun by bed (today and mid-June), soil, water, existing trees and plants, prioritized findings and recommendations, a seasonal action plan from your frost dates, materials with rough cost ranges, and methods and limits. "Print or save as PDF" prints just the report.
- Lot outline: an irregular lot plotted from your map is now one outline you can select

(Outlines → your lot) and reshape by dragging its corner squares, like the front and back yard. Typing a width or depth stretches it. No more second, fixed outline.

- At a glance: every tab can be moved on its own with ▲ ▼, and the ⇓ button moves a tab under the map (it opens there; ⇑ sends it back). Positions are remembered.
- Cleaner tabs: Water flow & drainage, Outlines, Measure objects, Where you're growing, Weather (with frost dates), Sun & shade, Soil health, Plants, House photos and Garden notes use section headings, cards and large buttons.
- Trees, bushes & buildings panel: preview map with colored dots, How it works, Why this matters, a Show filter, thumbnails cut from your photo, where each one sits ("in the front yard, near the street"), Tree/Bush/Building/Not an object buttons, Show more, Missed anything? and Dismiss all.
- "Show everything" now fits the whole top photo (or lot map), not just the lot.
- 3D: drag plants, trees, beds, rocks, mulch, barrels, decks and patios to move them; balconies stay on their wall and floor; downspouts slide along the house and bring their rain barrel. Plants dropped on a bed join it. Dragging empty space still turns the view.
- Setup restyled: numbered progress bar, tile grid with a summary, Growing area card, top-view setup cards, Photos & home orientation with quick tips and North/East/South/West buttons, Measure your house and yard with tips, and a card-style Review.
- Site reports are saved (on this device) every time you make one: reopen, reprint or delete them, export a backup or a CSV spreadsheet, import a backup, and see analytics (reports by month, common findings, ZIP codes, cities, average lot, sun mix, typical costs). Consultants can sign in to sync reports across devices; each report is encrypted on the device with a key protected by your passphrase before upload (Supabase). Addresses are never saved. A recovery key is shown once at sign-up.
- Plain-language help: tap any dotted-underlined garden word (about 80, like pH, mulch, full sun, frost date) for its meaning, why it matters and an example. A "? Help" button opens a searchable list of all garden words. Each At a glance section starts with a short beginner tip (can be turned off). Site reports end with a "Garden words used in this report" list.
- Privacy policy rewritten to match the app (version draft-0.5): a short summary, exactly what is saved in the browser, a card for each outside service (Open-Meteo, OpenStreetMap, Esri, jsDelivr, Supabase) saying when it is contacted and what it gets, permissions never asked for, and your choices. Turning saving off now deletes everything the site stored in the browser (with a warning first if saved reports would be lost); the sync sign-in is kept across visits only when saving is on. The site asks for your privacy choice again because the policy changed.
- One set of compass instructions: setup, Overview and the map all use the same five steps (go outside, hold the phone flat with the top edge pointing at the street, stay clear

of metal, wait for a steady reading, set and lock your front), the same Compass app advice (hold it flat too; turn on True north) and the same "Reading looks wrong?" fixes (figure 8, check with the sun, or skip the compass). The old instruction to hold the phone up for the Compass app is gone.

- Honest precision: modeled sun hours are rounded to the nearest half hour ("about 5 h", not "5.1 h"); modeled daily light (DLI) is a whole number; water amounts use two significant figures ("about 490 gal"); measured heights say "give or take" about 10%. Frost dates from weather records now show the range seen in most years (for example "between Apr 3 and Apr 19") and when it is safe to plant tender crops; without weather records they are labeled a rough guess from latitude that can be off by 2 to 3 weeks. The zone is labeled an estimate, with the USDA map as the official source. Site reports explain the rounding and where frost dates came from.
- Townhomes: maps usually draw a whole row of townhomes as one building. After tracing the building, "Split the row" asks how many homes are in it (a guess comes from the building length), draws numbered dividers, and you tap your own home. Homes of different widths can still be outlined by tapping corners. Tapping a house number or label on the map no longer traces just the label: the app starts from the building color nearby.
- Clear day vs real weather: every sun-hours number carries a "Clear day" label, and numbers that use recorded or forecast sunshine carry a "Real weather" label. The Sun & shade tab explains the difference and, once live weather is loaded, says what share of clear-day sunshine this week's forecast brings.
- Sources: a Sources page (footer) lists where each rule of thumb comes from (Penn State, University of Minnesota, Clemson, NC State, Michigan State and University of Nebraska–Lincoln Extension, NOAA, USDA, Open-Meteo, 811, ASPCA). Advice shows a small "source" link, and site reports end with a sources list.
- Care & watering (new At a glance tab): how much to water each bed this week (1 inch a week minus rain from your gauge or the forecast, more in hot weeks, less with mulch, gallons per bed, pots checked daily); a care plan for this month and next built from your plants and frost dates, each with "+ Task"; check-ins for how each plant is doing (doing well, okay, struggling, died, harvested); a harvest log; rough plant prices.
- Site reports price "Plants to buy" from rough garden-center ranges instead of "varies".
- Safety & wildlife (new tab): before-you-dig checklist (811, landlord or HOA, lead test, overhead wires); plants toxic to pets; invasive and fast-spreading plants; animals you see (deer, rabbits, groundhogs, squirrels, birds) with the plants they're likely to eat and the fence that stops them. Missing 811 calls, toxic or invasive plants and lead tests show up as report findings.
- Site conditions (new tab): windy spot, steep slope, road salt or sea spray, reflected heat, black walnut nearby, frost pocket, shade that will grow. Each adds advice and a report

finding.

- Model check: the Sun & shade tab shows how often your sunny and shady marks agree with the map.
- Quick height guesses: "Fill in quick guesses" in Measure objects gives every tree, bush, building and fence without a height a guess from its size or floors, labeled "quick guess" until measured.
- Straighten automatically: one button turns the top photo so its straight edges (roofs, fences, sidewalks) line up with your house outline.
- Better detection: found objects are marked "likely real" or "not sure" and sorted with the likely ones first. Find Fewer / Normal / More changes how sensitive the scan is.
- Tree line by direction: in Sun & shade, set how high distant trees, hills or buildings rise toward each of 8 directions (a fist at arm's length is about 10°). The shade model uses it.
- Español (beta): the ES button in the top bar switches menus, section names, beginner tips, consent choices and key steps to Spanish. Detailed text is still in English.
- Fixed: the home page's "Today's tip" had stopped showing after the plain-language help update.
- Your situation (new At a glance tab): pick where the garden is (home, community garden plot, school garden, rooftop, indoors or hydroponics) and tick what applies (kneeling is hard, wheelchair or seat, weak grip, low vision, tight budget). Advice adapts: raised and table beds, beds too wide to reach, 36-inch paths, drip lines and padded tools, high contrast, growing for less, community plot rules, school-year planting, rooftop weight and wind, and a grow-light calculator (PPFD × hours → daily light) for indoor growing.
- Climate-aware advice: hot and dry, hot and humid, tropical and short-season climates each get their own planting and heat tips, worked out from your region and zone.
- Android compass: when a browser reports turning but not true north, the app says so and suggests Chrome or typing the number from the Compass app. The compass steps mention calibrating with Google Maps on Android.
- Offline: on the published site, the app keeps a copy of its own files (sw.js, manifest.webmanifest, icon.svg) so it opens without internet after the first visit and can be added to a phone's home screen. No garden data is in that copy; the privacy policy says so.
- Client consent: the site report panel records whether the client agreed, how (in person, email, text, signed form) and when. Reports show the consent line.
- Labor and invoices: set an hourly rate to add a labor estimate to the report; "Make an invoice" opens an editable, printable invoice with labor and materials.
- Share reports: "Download file for the client" saves the report as a standalone page; "Email summary" opens your email app with the top recommendations.

- Test the sync service: under Saved reports, a step-by-step check that the live Supabase sync can be reached, read, written and cleaned up.
- Alerts: links to official sources (U.S. Drought Monitor, USDA APHIS, extension directory, invasive species lists) until verified alerts exist.
- Calendar reminders: "Add my tasks to my calendar" saves an .ics file with a morning reminder for each task. No notification permission needed.
- Ask about your garden (new tab): plain questions about frost dates, watering, planting times, sun in a bed, what grows where, garden words and pet safety, answered from your map on your device.
- Share cards: a picture of your map and plants with no address, plus a suggested image description copied for alt text.
- Fixed: tapping a house on the house map could trace the yard between property lines instead of the house. The app now tries the spots around your tap and keeps the smallest shape that doesn't run off the picture, which is the building. Tapping on a line or the house number works too.
- One set of measurements: tracing your house on the house map (or the lot map) fills in the house width and depth right away. Whatever you measure or type last, in any of the three ways, shows everywhere; typing a new lot width or depth reshapes a measured irregular lot. An outline too big for one house is flagged instead of filling the boxes.
- Measuring on the top photo: a note explains that photos rarely show property lines and offers the map screenshot instead. If the lot is already measured on the map screenshot, it says so and points to "Line up your top photo". The map screenshot can be shown beside the photo for reference.
- Tap the corners myself: on the house map and the lot map, "✎ Tap the corners myself" lets you outline the house (or building) by tapping each roof corner, then point 1 again or Done. Use it whenever the automatic outline doesn't fit; the size boxes fill in from your taps.
- Better automatic house outline on light map styles, where a house is only a shade darker than the yard: if the first try spreads into the yard, the app looks again with a tighter color match. An outline bigger than about 110 ft or 5,000 sq ft is treated as the yard and not used.
- Fixed: Identified objects could show an empty list. The tab now appears whenever there's a top photo, with "Scan my photo" and "Find more" when nothing has been found; a filter that hides everything falls back to showing all; "Find fewer" can't make the list disappear; and the tab itself shows a quick list where you can mark each item as tree, bush, building or not an object.
- Problems & plan (new At a glance tab): every item on the map is labeled Existing or Recommended. "New things I add are" switches between them, and each item's card

has an "On the plan" picker. Mark problems by tapping the map (standing water, erosion, bare soil, sick plants, invasives, too much shade, downspouts, pests, weeds, safety hazards), with notes and a Fixed checkbox. The map switches between Today, With recommendations and Both, with a legend: dashed green for recommended, red × for removals, red numbered pins for problems, green ✓ for fixed. Recommended items have Done / Planted / Removed buttons.

- Site reports show two maps side by side ("Site today" and "With recommendations"), and each open problem is a numbered finding that matches its pin.
- Visits over time: every saved report keeps a copy of the map plan (never the address). Step through visits with a slider, or compare two visits (or a visit and now) on the map: green is new, red × was removed, ✓ marks fixed problems, with a short summary. Reports include "Progress since the last visit": each recommendation and problem from the last visit marked done, not started, fixed or still open, plus new problems.
- Privacy policy updated (draft-0.7) to say saved reports include a copy of the map plan.
- Regenerative gardening is now a core focus. New "Regenerative practices" tab: a soil health score built on the USDA NRCS principles (keep soil covered, disturb it less, keep living roots, grow many kinds of plants, feed pollinators, close the loop, keep rain on site), each with where you are now, the next step and "+ Task"; a checklist of what you already do (no-dig, compost, keep leaves, chop and drop, cover crops, no synthetics, habitat); and cover crop timing from your frost dates. The care plan adds cover crop, leaf and no-dig compost steps; site reports include a soil health scorecard; new plain-language words: regenerative gardening, no-dig, cover crop, chop and drop, living roots. The welcome page mentions building healthy soil.
- Yard conditions (new At a glance tab): slope arrows (which way is downhill and how steep, with a drop-over-distance helper), height pins (higher or lower than the house) and dips, drawn as height lines on the 2D map and as real ground in 3D, with a Heights ×1/×2/×3 and Color by height option to make slopes and dips easy to see. Add tree stumps (with a plan: keep, rot in place, grind, dig out), retaining walls (block, timber, concrete or boulders; tall walls flag an engineer or permit), stairs (stone steps set into the slope, wood or concrete; step count and riser height from your slope), walkways and driveways.
- Where water goes: water lines run downhill from your slope, downspouts and hard surfaces; lines that reach the house turn red, likely puddles show in blue, and green dashed lines suggest where it should go instead (around the house with a swale or French drain, or to a rain garden). Draw your own "where water should go" routes. Show water moving and rain falling on the 2D map and in 3D (still arrows when reduced motion is on).
- Ways to water (new At a glance tab): this week's rain vs. about 1 inch a week, and for each bed which of rain, hose, sprinkler, PVC line or watering can reaches it and how

many gallons it needs beyond the rain. Add a hose and spigot (with hose length and reach circle), sprinklers (radius and pattern), watering cans and rain barrels.

- PVC pipes: tap points to lay a pipe; corners become elbows, a pipe that starts on another pipe gets a tee, and an end next to a spigot, rain barrel or downspout connects to it. Each pipe lists its parts (ten-foot lengths, elbows, tees, couplings, adapters, end caps). Use it to water beds or to carry water, add branches and extend it, and turn a suggested water route into a pipe.
- Photos from different angles: add ground-level photos from a folder (never the whole photo library; location data removed), describe them, set which way the camera faced and where you stood, then mark stumps, dips, walls, steps, spigots, bamboo or ivy from the photo.
- - Bush works like + Tree. Running bamboo and English ivy can be marked as problems, show up in 3D, and come with a removal plan (University of Maryland Extension) in the plan and report. Muscadine grape and crimson clover are in the plant library; a trellis arch can grow grapes, beans, cucumbers, roses, clematis or coral honeysuckle.
- 3D uses realistic free models (credited on the Sources page and in the README): trees chosen by evergreen, columnar, fruit or shade type, bushes, bamboo, banana, raised beds, greenhouse with seedlings, shed, deck, balcony railings, rain barrel, hose, watering can, walls, stairs, stone walkways, crops and dirt rows, grapes, ivy and dandelions.
- Easier 3D navigation: drag to turn and tilt; two fingers pinch to zoom, twist to turn and slide to move; double-tap zooms in where you tap; scroll zooms toward the pointer; zoom, turn, tilt and reset buttons; Shift + arrow keys move the view and Home resets it.
- Site reports add a "Yard conditions and water" section (slope, features, where water goes now and where it should go, ways to water each bed, PVC parts) and flag water running toward the house.
- Budget first. New "Plan on a budget" tab (Start With What You Have): pick where you grow, a spending limit ($0, $25, $50, $100 or your own amount), tap what you already have and what your household eats (tap twice for foods you don't). The plan picks crops that fit your sun, season and space, with planting dates from your frost dates, then lists what you already have, what to buy within the limit (with rough prices and why), and what to find free, reuse, borrow or wait on, with food-safety notes from UMD Extension. A $0 plan uses only what you have and free options. "Use this plan" adds the purchases to your budget, marks them Needed in your supplies and adds planting tasks. The app never asks about income.
- New "Supplies & budget" tab: one budget for the year (limit, spent including receipts, planned, money left, and what you saved by borrowing, reusing or getting things free). Mark each item Planned, Bought, Got it free or Borrowed and type what you paid.

Supplies on hand are Owned, Borrowed (with who lent it), Reused or Needed. "Before you buy" shows what you already have, what to reuse or borrow, what to check first (soil test, pest ID) and safety notes.

- My next three tasks, at the top of At a glance and on Home: the three things that matter most now, from your tasks, frost and dry spells, problems, planting dates, harvests and this month's care. Done, Later (3 days) or Skip (a week). A Quick / Detailed switch keeps extra record fields optional.
- The harvest log is now its own At a glance tab: log what you picked and where it went (eaten fresh, stored, given away, traded). Stored food shows as on hand and can be marked used. Detailed mode adds the date, bed, notes and an optional store-price comparison with what you spent.
- Privacy policy updated (draft-0.8) to list the spending limit, supplies list and harvest log among the data kept on your device.
- Accounts for customers and the consultant (admin), synced across phone, tablet and computer through Supabase. Customers see a simpler app: the same dashboard plus Sun, shade & water views with a time slider, Where you're growing, 3D, Water flow & drainage, Care & watering, Costs & receipts (photo or typed), Report a problem, Questions, Journal and What you grew. Their map and 3D are view-only and At a glance is hidden. The garden tab shows the customer's name ("Jordan's Garden").
- The consultant sees every customer's garden, with a Customers list (unread messages, new entries), Add a customer (set up their yard without touching your own garden, or copy the garden on screen), invite links by email, text or copy, a QR code people scan to say they're interested, the interest list, and a "Customer: messages & reports" tab for replies and turning reports into map pins.
- Explore is now the Growing Circle: account holders start or join circles by ZIP code, pick crops to grow together, say what each person grows, and post updates, extras to share and requests. Circle leads and the admin can hide posts and remove members; anyone can report a post; posts with street addresses or phone numbers are blocked.
- New `supabase-setup.sql` with the tables, row-level security, invite and flag functions, and private storage for receipt and map photos. Privacy policy updated (draft-0.9) to describe garden-service accounts and the Growing Circle.
- No more passwords: sign in or create an account with a 6-digit code sent by email (or by text once a texting service is connected). An authenticator app can be added as a second step from Your account; the admin account must use one, and the database only grants admin access after that step. Saved consultant reports keep syncing; older passphrase-locked reports move over after typing the old passphrase once. Privacy policy updated (draft-0.10).

Oct 27, 5:30 PM — Trees that look like real trees

- New tree shape: "Woods tree: very tall, leaves only near the top" (leaves start about two-thirds of the way up). More tree types: American beech, hickory, black gum, black cherry, ash, and tulip poplar in the woods.
- "Top of the tree" for each tree: full of leaves, a few bare branches, top third bare, or half or more bare. Bare tops let more sun through, and the 3D view shows them as gray dead branches. "Vines climbing the trunk" shows ivy on the 3D trunk.
- 3D trees are drawn from each tree's type and shape: trunk height to the first leaves, crown form (round, oval, wide, vase, cone, column, weeping, woods), bark color, spring bloom, and fall color by species (red maples red, tulip poplars yellow, oaks russet, sweetgums red and orange). Leaves fill in, change color and drop with your frost dates, and winter shows bare branches.
- Woods outlines: "What the trees look like" (very tall with leaves near the top, the new default; leaves from halfway up; or full) and "Trees with bare or dead tops." Woods in 3D are a mix of tulip poplar, oak, red maple, sweetgum, hickory, beech and black gum, with the same fall colors.
- Tree health check (from a tree's panel or Calendar & journal → Tree health): check what you see (hanging branches, cracks, new lean, fungus, dead top, holes or sawdust, cavities, bark, vines, girdling roots, leaf problems and more) and get Watch it, Act this season, or Call someone now, with likely causes, what to do, species warnings (emerald ash borer, oak wilt, Callery pear) and the arborist stop point. Results go in the journal, add a follow-up task, and put a ! marker on the 2D map.

Oct 27, 6:30 AM — Round 2: small versions for the pilot

- Important-date alerts: a bold banner at the top of Home when a key date arrives for this garden (feeding, pruning, starting seeds, first and last frost, covering tender plants, cool- and warm-season planting, bulbs and garlic, emptying the rain barrel). Each has Done (noted in the journal), Remind me in 3 days, Don't show alerts like this, and an × to close. More dates fold under "N more dates". Screen readers hear the top one once. The same dates appear in My next three tasks, and Done there clears the alert. Profile → Garden date alerts turns each kind on or off. Nothing is emailed or texted.
- Feeding and pruning calendar (My Garden → Calendar & journal): feeding dates for each crop from its planting date (or "no feeding needed" with the reason), a fertilizer calculator (first bag number and area → cups or tablespoons, conservative without a soil test), "Don't overdo it, and homemade fertilizer", pruning windows for every tree and shrub on the map and in beds (oaks only late fall to late winter, spring bloomers right after flowering, and more), 3 common cuts illustrated, and the arborist stop point.

Neighbors' trees are left out; street trees get a city-forester note.

- Plant library weather groups: Cold weather, Mild or either, and Hot weather filters that combine with the other filters. Each plant shows its planting dates from your frost dates ("Plant Oct 7 – Nov 11 (fall)"), "Plant now" when it's time, and heat notes for crops that bolt. New crops: spinach, broccoli, collard greens, radish, beets, carrots, green onions, okra, pansies and sunflower.
- Expected harvest (My Garden → Beds & plants): poor, decent and optimal pounds for each crop in each bed, with the likely range highlighted from the bed's sun (midsummer for warm crops, spring for cool ones), drainage and soil pH, plus what's been logged this year and a whole-garden total. Marked "Draft ranges" until the consultant reviews the numbers.
- Customer Details (consultant): a Project section with stage (draft, client review, approved, scheduled, completed), site visit, estimate, estimate sent, an hours log, and invoice sent. Customers list shows each stage and hours; the Excel and PDF lists add project columns; the dashboard has a Projects card (count by stage, hours this month, complete records, average days from visit to estimate, finished projects with no invoice date).

Oct 27, 7:30 AM — Round 1: make the app match real yards

- Shade accuracy: each tree's panel has "What does it look like?" picture buttons (tall bare trunk, full to the ground, wide, thin top, column, evergreen cone, fruit tree, bush, bare) and "Or pick a tree type" (25 common trees). They set where the leaves start, how full the tree is, its leaf season and its 3D model. A "Lowest branches" box lets sun pass under high crowns.
- Leaves fill in over about 4 weeks in spring and thin over about 5 weeks in fall, instead of switching on and off.
- Right now shows how much of each bed is shaded (for example "Partly shaded · 40%") and which tree or building is shading it. A bed counts as in sun unless at least 15% is shaded.
- Compass: a locked compass front stays put when "Street is on another side" is used; the map compass label shows the saved front ("Saved: front faces south (179°) from your Compass app · tap for live"); typed Compass-app numbers ask "True north / Magnetic north / Not sure" and convert magnetic readings for your area.
- Raised beds in 3D are built from boards to each bed's exact outline, with the soil inside the frame (the old model was two tilted beds with a solid top).
- Greenhouses are fixed sizes: kit sizes from 6 × 4 to 12 × 20 ft, or type length, width and height. Dragging moves them; it no longer stretches them.
- Turn outlines by 1°, 5° or 15°, or type an angle; "Fit to the building in the photo" snaps a

house, shed or neighbor's building to a building found in the top photo.

- Move without dragging: "Tap where it goes" and arrow buttons for objects and single corners on the map and on the setup screens.
- Text sign-in: "We couldn't send a text right now. Use your email instead" when Twilio refuses to send; "isn't switched on yet" only when it's actually off.
- 3D ground: brown height lines and blue water lines (green for suggested routes) drawn on the 3D ground, with Height lines and Water lines buttons; nearly flat yards show heights 3 times taller automatically, with a note.

Oct 22, 9:00 PM

- Woods: draw around a wooded area and the app fills it with trees (close together, medium or spread out; height; leafy, evergreen or mixed). They cast shade (bare in winter unless evergreen), show in 3D, and refill when you change the outline.
- Ponds (draw around them) and streams, creeks, rivers or drainage ditches (draw along them, with width and flow direction). Shown in blue on the map and in 3D, with depth, "all year or wet seasons only", and notes.
- Hill or mound and Uneven ground: tap to place, set the height or how bumpy, drag corners to size. The 3D ground and height lines show them.
- Yard conditions (At a glance) lists woods, water and ground shapes with buttons to add each. The outline editor works on woods, ponds and streams too.

Oct 22, 7:45 PM

- My Garden map: edit the lot and growing-place outlines. Pick one with the new "Edit an outline" buttons (or tap inside a growing place or on an outline), then drag a corner, tap + between corners to add one, or tap a corner and press "Remove corner". Lot size and area update as you go; Undo works.

Oct 22, 7:15 PM

- "Account" button in the top bar of every garden screen (settings and Sign out) for anyone signed in.

Oct 22, 7:00 PM

- Delete a customer's garden: Customers → Details → Delete this garden (removes its map, journal, harvests, receipts, reports, messages and invite).
- If the server quietly refuses a change (usually a sign-in that lost its authenticator step),

the app now says "Not saved" and explains how to fix it instead of showing "Synced".
- Fix: after signing out and back in, the customer list could come back empty. Signing out now resets the sync.

Oct 22, 6:00 PM

- Setup maps (house map, lot map, top photo): zoom in and out with – / + / Fit, pinch with two fingers (buttons only on the top photo), or Ctrl + scroll on a computer.
- Edit outline corners while plotting or after: tap a corner to select it, drag it, remove it, or delete all of that outline's corners; tap + between corners to add one you missed. Arrow keys nudge a selected corner; Delete removes it. Undo covers every edit.
- "Clear all lines" is now "Delete all points".

Oct 22, 4:30 PM

- Consultant dashboard: the admin lands on it after signing in (overview tiles, needs your attention, review queue, community impact; links to Customers, Gardens, Growing Circle and Reports). "Dashboard" button next to Customers returns to it.
- Customer list download: Excel sheet or PDF from the Customers screen (name, contact, year of birth, ZIP, garden today, lot size, gardening space, needs, stage, signed-up and contacted dates).
- Interest form asks (all optional) for year of birth, garden today, what they need, lot size and gardening space. Under-14 years ask a parent or guardian to fill it in.
- "Contacted today" button on interest requests; "Details" editor for each customer.
- "Start my garden" skips the account step when already signed in. Clearer message when the code limit is reached.
- Database: run section 6b of supabase-setup.sql once.